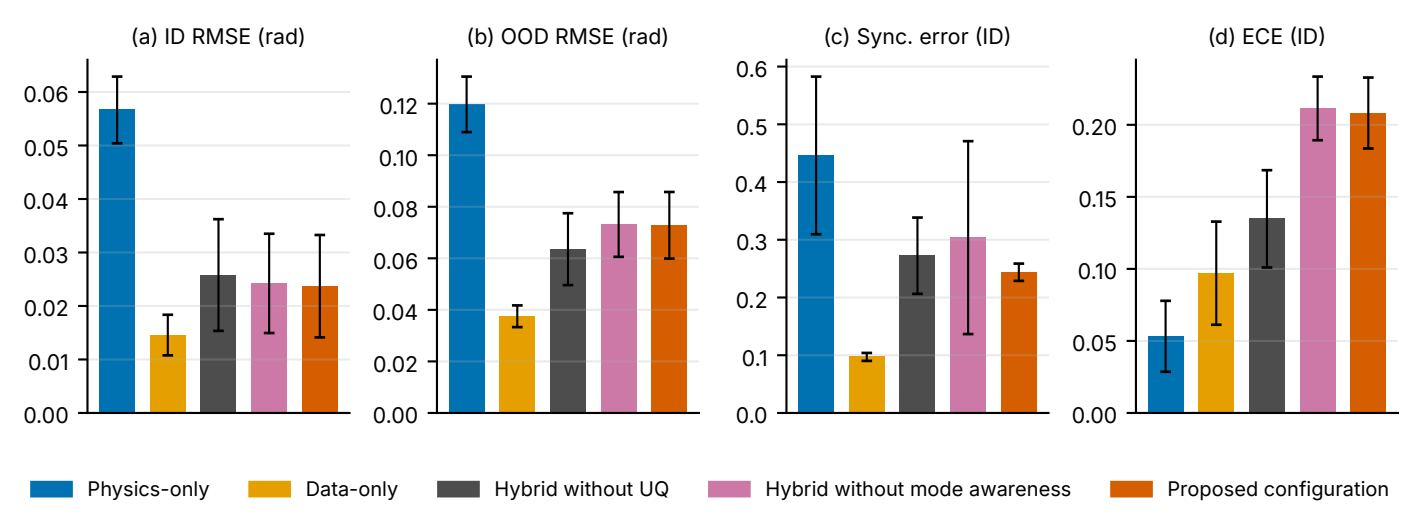
(a) ID RMSE (rad)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
(b) OOD RMSE (rad)
0.12
0.10
0.08
0.06
0.04
0.02
0.00
(c) Sync. error (ID)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
(d) ECE (ID)
0.20
0.15
0.10
0.05
0.00
Physics-only
Data-only
Hybrid without UQ
Hybrid without mode awareness
Proposed configuration